

CV5570 Assignment 2 - Object Detection Report

Indian Institute of Technology Madras

CV5570: Computer Vision for Construction Engineering

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GitHub repository: <https://github.com/wick1-22/Assignment-2-CV5570>

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1. Summary

This report compares Faster R-CNN, RetinaNet, and RT-DETR-L on 3,929 ACID images. The highest validation mAP@0.5:0.95 was from RT-DETR-L (0.750); the highest measured speed was RetinaNet (16.8 FPS). All results use the same saved validation split.

2. Objective and Scope

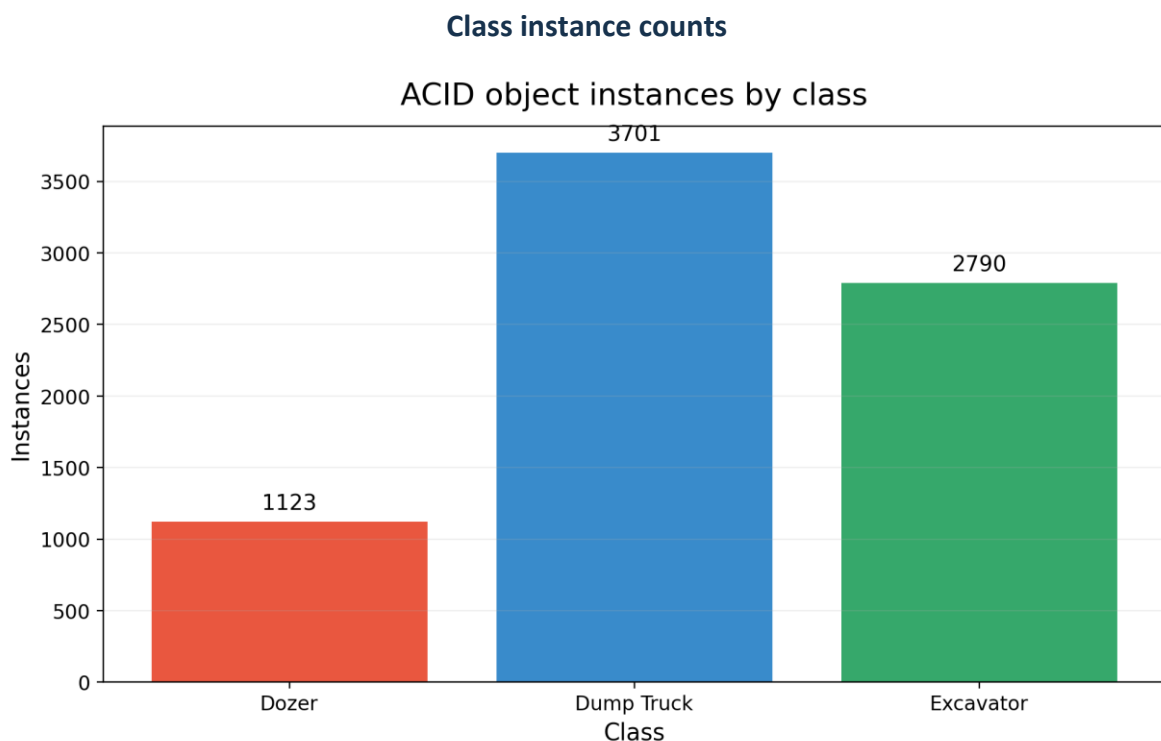
The aim was to train one detector from each of the two-stage, one-stage, and transformer families on the ACID dataset. The comparison reports class-wise average precision, speed, parameter count, training time, loss history, and validation examples for Dozer, Dump Truck, and Excavator.

3. Dataset Description and Preprocessing

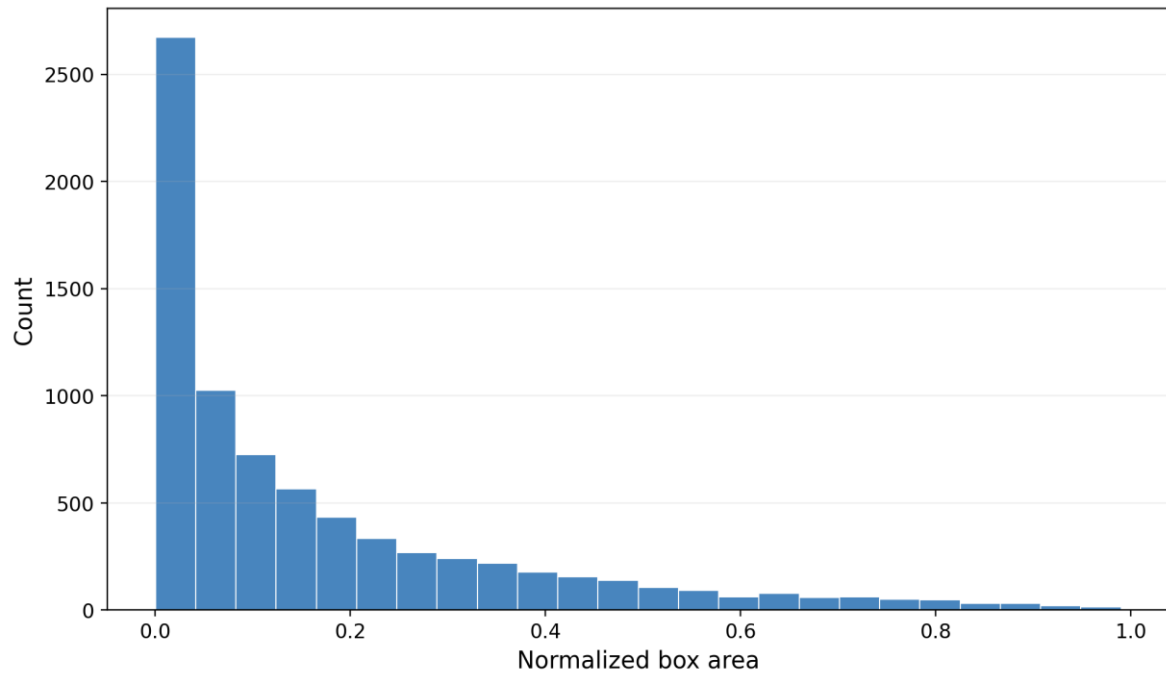
The selected subset contains 3,929 images and 7,614 object instances. Annotations are in COCO JSON format with pixel boxes stored as [x, y, width, height]. The split uses seed 42 and proportions 70/20/10.

Split	Images	Instances	Dozer	Dump Truck	Excavator
Train	2,750	5,351	776	2,614	1,961
Val	785	1,499	243	715	541
Test	394	764	104	372	288

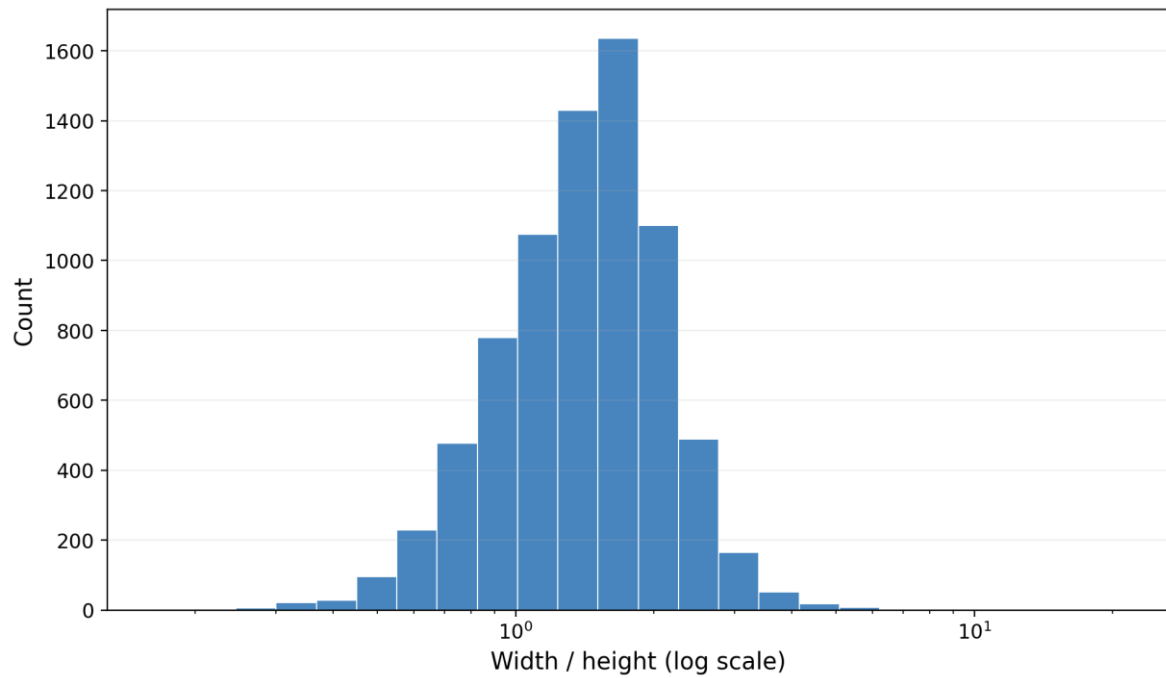
The median normalized box area is 0.088; the median width-to-height ratio is 1.45. The held-out test split is not used for the reported model-selection metrics.



Bounding-box area as a fraction of image area



Bounding-box width-to-height ratio



Six ACID images with ground-truth boxes

Ground-truth sample 1



Ground-truth sample 2



Ground-truth sample 3



Ground-truth sample 4



Ground-truth sample 5



Ground-truth sample 6



4. Detector Methods and Training Setup

Faster R-CNN: The region proposal network selects candidate regions; the second stage classifies and refines them. It was chosen as a two-stage baseline for equipment that appears at different sizes in cluttered construction scenes (Ren et al., 2015). The model starts with TorchVision COCO-pretrained ResNet-50-FPN v2 weights.

RetinaNet: Predicts class scores and box offsets in one pass. Its focal loss reduces the contribution from easy background locations, which are common in object detection (Lin et al., 2017). It provides the one-stage comparison. The model starts with TorchVision COCO-pretrained ResNet-50-FPN v2 weights.

RT-DETR-L: This transformer detector uses an encoder/decoder and set-based predictions. It was selected to compare a transformer-based approach on cluttered scenes and to measure its speed/accuracy trade-off (Zhao et al., 2024). It starts from the pretrained `rtdetr-l` checkpoint. The prediction interface used here does not return attention maps, so its errors are reviewed from the validation boxes.

The models use the same saved split, 640-pixel inputs, 10-epoch budget, and disabled augmentation. The model-specific optimizer settings and runtime values are recorded below.

Model	Checkpoint	Optimizer	Learning rate	Batch	Epochs	Device
Faster R-CNN	Faster R-CNN ResNet50-FPN v2 COCO weights	SGD	0.0050	2	10	CUDA GPU
RetinaNet	RetinaNet ResNet50-FPN v2 COCO weights	SGD	0.0005	2	10	CUDA GPU
RT-DETR-L	<code>rtdetr-l.pt</code>	AdamW	0.0014	2	10	CUDA GPU

Faster R-CNN used SGD at learning rate of 0.005. RetinaNet used 0.0005 after the initial 0.005 run produced non-finite losses. Both used momentum 0.9 and weight decay 0.0005. RT-DETR used the optimizer settings saved by Ultralytics.

5. Validation Results

Model	Family	mAP@0.5	mAP@0.5:0.95	FPS	Parameters (M)	Training time (min)
Faster R-CNN	two-stage	0.929	0.679	9.6	43.27	138.7
RetinaNet	one-stage	0.896	0.625	16.8	36.37	79.3
RT-DETR-L	transformer	0.939	0.750	13.0	32.81	72.3

Class-wise average precision

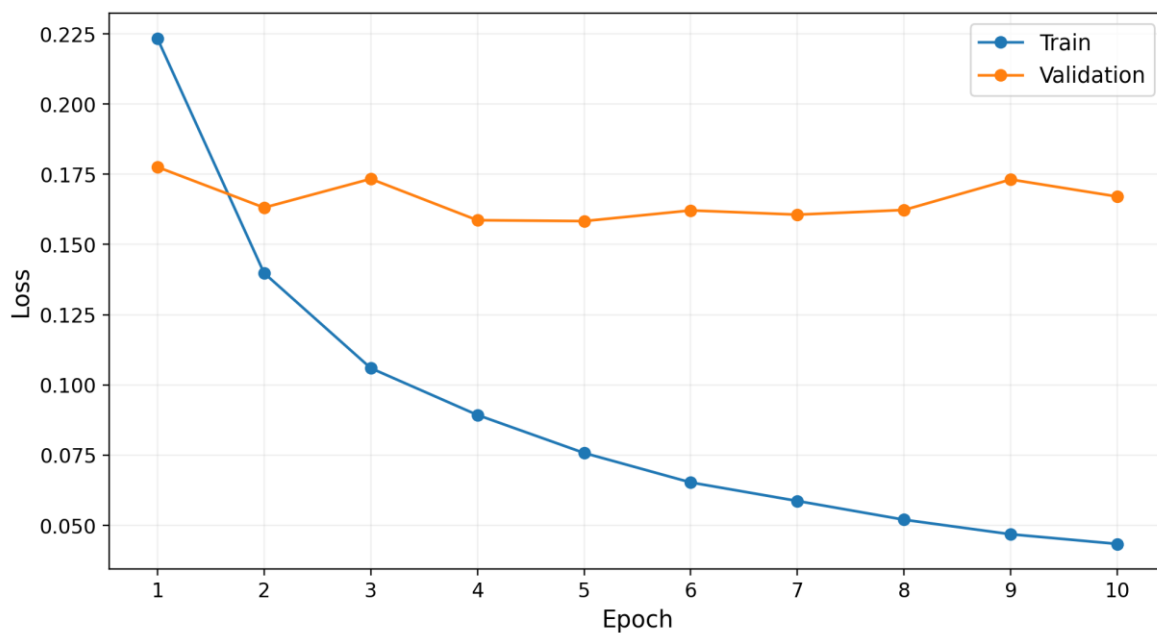
Model	Class	AP@0.5	AP@0.5:0.95
Faster R-CNN	Dozer	0.937	0.691
Faster R-CNN	Dump Truck	0.896	0.617
Faster R-CNN	Excavator	0.953	0.729
RetinaNet	Dozer	0.939	0.674
RetinaNet	Dump Truck	0.817	0.549
RetinaNet	Excavator	0.933	0.654
RT-DETR-L	Dozer	0.948	0.754
RT-DETR-L	Dump Truck	0.910	0.677
RT-DETR-L	Excavator	0.958	0.818

6. Training and Convergence

Faster R-CNN: its validation objective changed by 5.9% from epoch 1 to the last epoch and was flattening or changing direction near the end. RetinaNet: its validation objective changed by 37.2% from epoch 1 to the last epoch and was still decreasing at the end. RT-DETR-L: its validation objective changed by 27.4% from epoch 1 to the last epoch and was still decreasing at the end. The minimum recorded validation loss occurred at epoch Faster R-CNN 5, RetinaNet 10, RT-DETR-L 10. For 10 epochs, recorded training time was Faster R-CNN 138.7 min, RetinaNet 79.3 min, RT-DETR-L 72.3 min. RT-DETR-L reached its lowest recorded validation loss at epoch 10, compared with epoch 5 for Faster R-CNN and epoch 10 for RetinaNet. Its 72.3-minute run was shorter in wall time than Faster R-CNN (138.7 min) and RetinaNet (79.3 min);

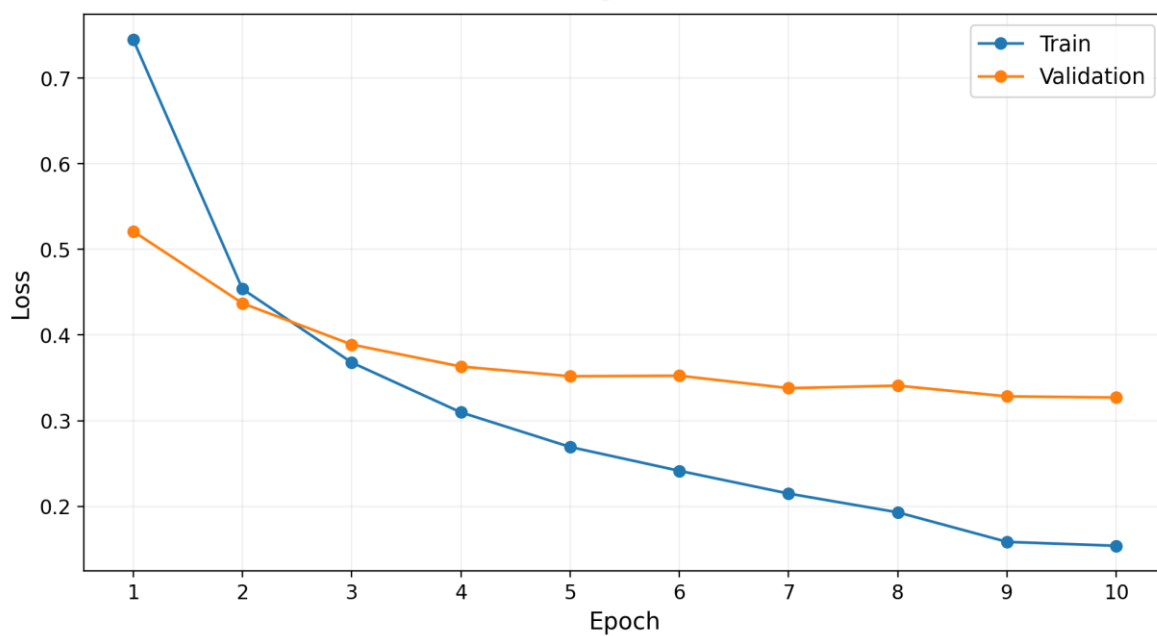
loss definitions differ, so this compares convergence timing rather than loss magnitude. Loss definitions differ across architectures, so compare their within-run trends and time-to-best rather than the absolute loss values.

Faster R-CNN training and validation loss

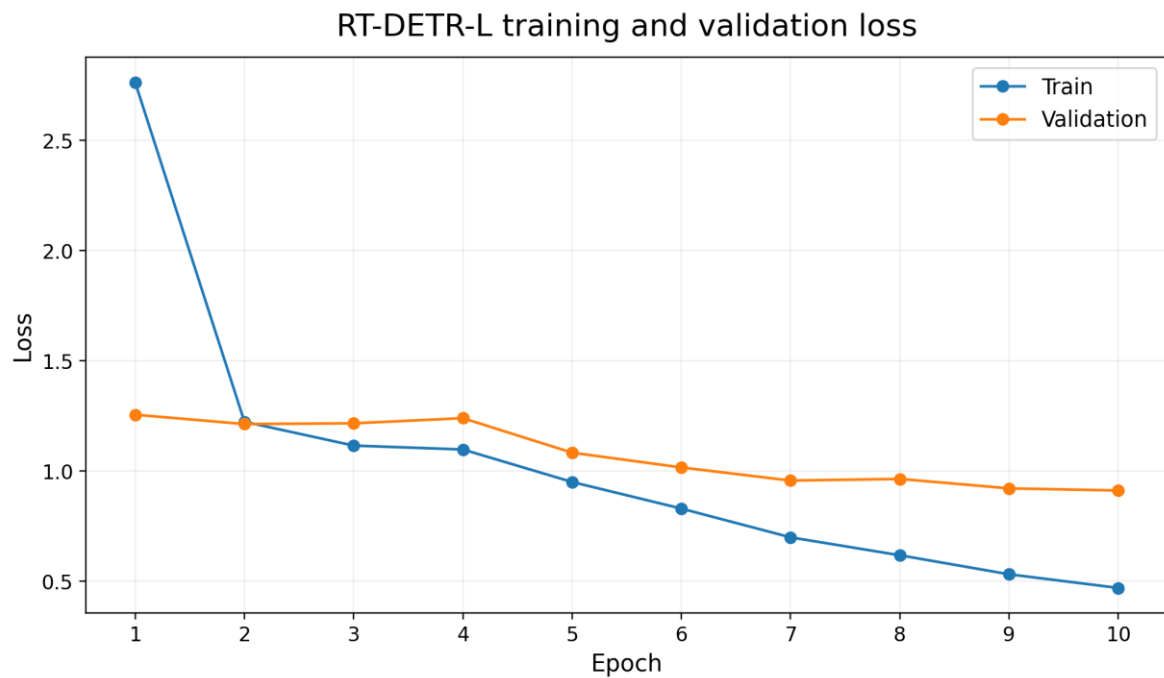


Faster R-CNN loss curves

RetinaNet training and validation loss



RetinaNet loss curves



RT-DETR-L loss curves

7. Detection Examples and Failure Analysis

Green boxes mark ground-truth objects and red boxes mark predictions. The boards show selected validation successes and failures using the same selection rule for each model.

Faster R-CNN

Success: 00002.jpg
TP 2 FP 2 FN 0



Success: 00018.jpg
TP 4 FP 2 FN 1



Success: 00028.jpg
TP 1 FP 1 FN 0



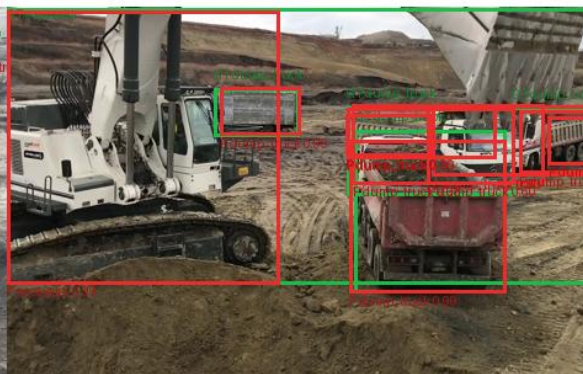
Failure: 04460.jpg
TP 6 FP 9 FN 2



Failure: 05513.jpg
TP 6 FP 7 FN 1



Failure: 09385.jpg
TP 5 FP 6 FN 1



Faster R-CNN validation detections

Type	Image	TP	FP	FN
Success	00002.jpg	2	2	0
Success	00018.jpg	4	2	1
Success	00028.jpg	1	1	0
Failure	04460.jpg	6	9	2
Failure	05513.jpg	6	7	1
Failure	09385.jpg	5	6	1

RetinaNet

Success: 00002.jpg
TP 2 FP 1 FN 0



Success: 00018.jpg
TP 4 FP 2 FN 1



Success: 00028.jpg
TP 1 FP 0 FN 0



Failure: 07480.jpg
TP 4 FP 10 FN 4



Failure: 08970.jpg
TP 6 FP 9 FN 2



Failure: 05513.jpg
TP 5 FP 9 FN 2



RetinaNet validation detections

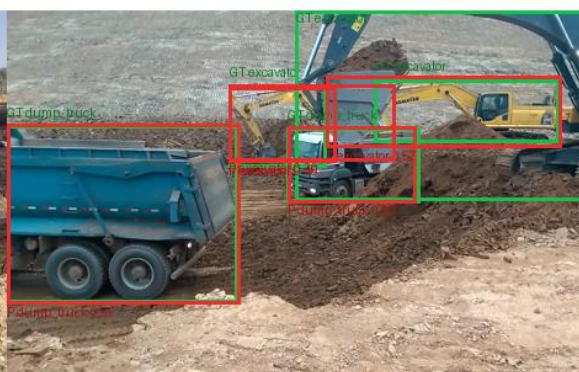
Type	Image	TP	FP	FN
Success	00002.jpg	2	1	0
Success	00018.jpg	4	2	1
Success	00028.jpg	1	0	0
Failure	07480.jpg	4	10	4
Failure	08970.jpg	6	9	2
Failure	05513.jpg	5	9	2

RT-DETR-L

Success: 00002.jpg
TP 2 FP 1 FN 0



Success: 00018.jpg
TP 4 FP 1 FN 1



Success: 00028.jpg
TP 1 FP 0 FN 0



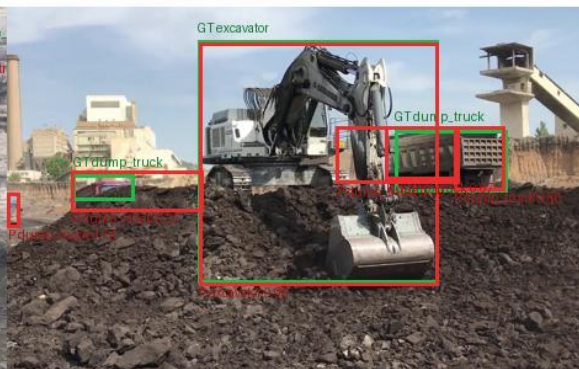
Failure: 06108.jpg
TP 0 FP 11 FN 1



Failure: 05513.jpg
TP 6 FP 7 FN 1



Failure: 02159.jpg
TP 1 FP 5 FN 2



RT-DETR-L validation detections

Type	Image	TP	FP	FN
Success	00002.jpg	2	1	0
Success	00018.jpg	4	1	1
Success	00028.jpg	1	0	0
Failure	06108.jpg	0	11	1
Failure	05513.jpg	6	7	1
Failure	02159.jpg	1	5	2

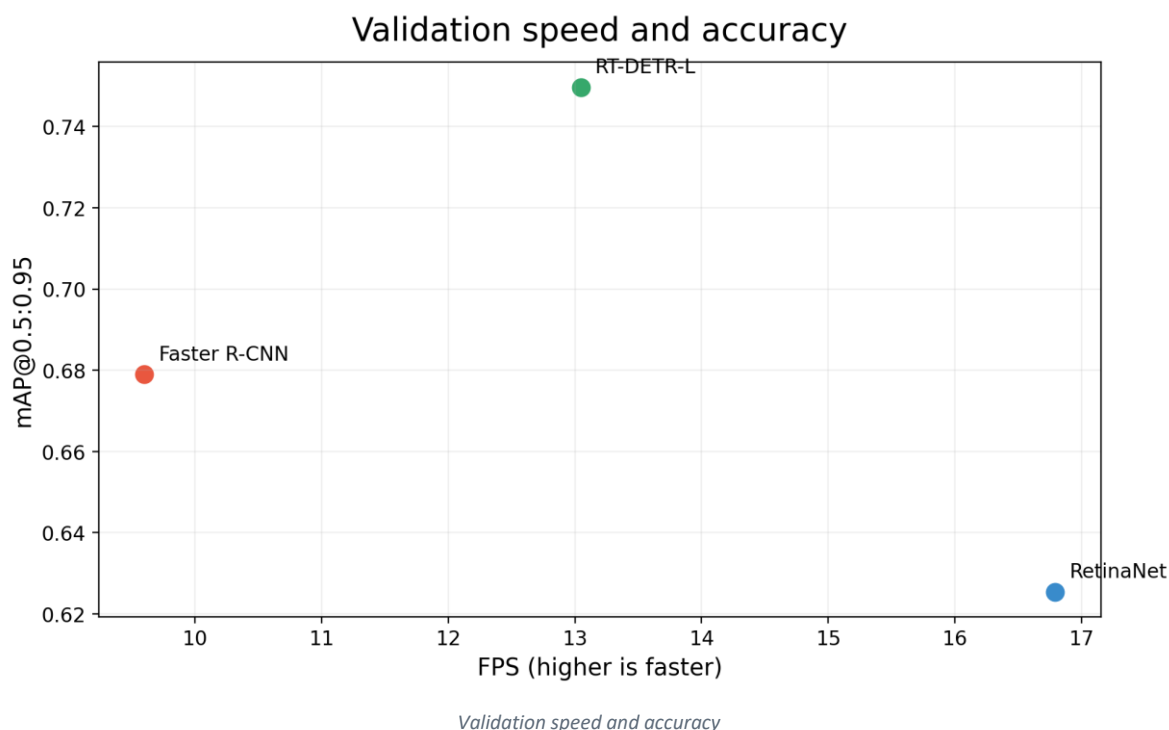
Observed in the displayed validation examples: Faster R-CNN detects the excavator in 00002.jpg (TP 2, FP 2, FN 0) and the dozer in 00028.jpg (TP 1, FP 1, FN 0), while the crowded or overlapping scenes 04460.jpg, 05513.jpg, and 09385.jpg contain extra predictions and missed objects. RetinaNet detects the isolated dozer in 00028.jpg, but the crowded scenes 07480.jpg and 08970.jpg contain several false boxes and missed objects. RT-DETR produces no correct match for

the dump truck in 06108.jpg (11 false positives and one missed object); the crowded scenes 05513.jpg and 02159.jpg also show overlapping extra boxes and missed instances.

Possible explanations to investigate: small objects at 640-pixel input resolution may be harder to localize; CNN anchor sizes may not fit the observed box shapes; the class distribution is imbalanced; and overlap or NMS may contribute to duplicate or suppressed detections. These are hypotheses, not causes established by this experiment.

8. Comparison and Deployment Discussion

The fastest measured model was RetinaNet at 16.8 FPS. The highest mAP@0.5:0.95 was from RT-DETR-L at 0.750. These results make the fastest model a candidate for real-time monitoring and the most accurate model a candidate for offline review, subject to the class-wise results and failure examples.



9. Limitations

The experiment uses one random image split and a short training budget. The split is not grouped by scene, so visually similar images may occur in different subsets. The detector families use different training implementations and optimization settings. The reported speed depends on the recorded GPU and inference code. Validation results do not replace evaluation on the held-out test split.

10. Conclusion

On this validation split, RT-DETR-L had the highest mAP@0.5:0.95 (0.750) and RetinaNet had the highest speed (16.8 FPS). The choice between them depends on whether accuracy or throughput is the main requirement. The class-wise scores and example failures should be considered before deployment.

11. References

1. Ren et al., 'Faster R-CNN: Towards Real-Time Object Detection with Region Proposal Networks,' NeurIPS, 2015.
https://papers.nips.cc/paper_files/paper/2015/hash/14bfa6bb14875e45bba028a21ed38046-Abstract.html
2. Lin et al., 'Focal Loss for Dense Object Detection,' ICCV, 2017.
https://openaccess.thecvf.com/content_ICCV_2017/papers/Lin_Focal_Loss_for_Dense_Object_Detection_ICCV_2017_paper.pdf
3. Zhao et al., 'DETRs Beat YOLOs on Real-time Object Detection,' CVPR, 2024.
https://openaccess.thecvf.com/content/CVPR2024/html/Zhao_DETRs_Beat_YOLOs_on_Real-time_Object_Detection_CVPR_2024_paper.html

4. ACID dataset, link provided in the assignment brief: https://drive.google.com/uc?id=1Qg_X5FygUMBRTcVFPb0s1fQP-20f8n8O
5. Ultralytics RT-DETR documentation: <https://docs.ultralytics.com/models/rtdetr>